Experiment 6: Online Cake Ordering System

Aim

To design and develop an online cake ordering website using HTML, CSS and JavaScript in which customers can register, log in, search and filter cakes by category, sub category (flavour) and price range, view cake descriptions, customise size and message, add cakes to a cart, manage shipping addresses, place orders and view order and shipping status; and the admin can post and manage cake details with images, manage orders, update order status and the shipping location.

Algorithm

1. Start and display the login page (Login and Registration). Registration validates name, email, 10-digit phone, password and a unique username.
2. Login: verify credentials for the selected role (Customer / Admin) and open the matching menu.
3. Customer - Cakes: read the search text, category, sub category and minimum / maximum price; display only the cakes that satisfy all the filters, each with image, description and price per kg.
4. Customer - Add to cart: select the size (0.5 kg, 1 kg, 2 kg) and an optional message; price = price per kg x size factor; add the item to the cart; remove items if needed.
5. Customer - Shipping address: add, update or delete addresses after validating the 6-digit PIN and 10-digit phone number.
6. Customer - Place order: select the address, delivery date (after today) and payment mode; save the order with status Placed, empty the cart and send the confirmation email.
7. Customer - My Orders: display each order with the total, delivery date, current status and shipping location.
8. Admin - Products: post a new cake (name, category, flavour, price, description, uploaded image) and edit or delete existing cakes.
9. Admin - Orders: view all orders with delivery details; update the status (Placed, Confirmed, Baking, Out for Delivery, Delivered) and the shipping location.
10. Logout returns to the login page. Stop.

Program (HTML, CSS and JavaScript) - cake.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Sweet Slice - Online Cake Shop</title>
<style>
:root{box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);
--bg:#fff5f7;--card:#fff;--tx:#3a2430;--mut:#8a6b77;--ac:#c2185b;--acx:#fff;--bd:#f0d3dc;--bad:#c0392b;--ok:#1e8449}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#1c1217;--card:#271a21;--tx:#f5e6ec;--mut:#b99aa7;--
    ac:#f06292;--acx:#2b0a15;--bd:#4a2f3a;--bad:#ef7b6e;--ok:#58c183}}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--tx);font-family:system-ui,Segoe UI,Arial,sans-serif;line-height:1.45}
header{background:var(--ac);color:var(--acx);padding:12px 16px}
header h1{margin:0 0 8px;font-size:1.2rem}
header nav{display:flex;flex-wrap:wrap;gap:4px}
header button{background:transparent;color:inherit;border:1px solid transparent;padding:5px 10px;border-
    radius:5px;cursor:pointer;font-size:.88rem}
header button.on,header button:hover{border-color:currentColor}
main{max-width:1000px;margin:16px auto;padding:0 12px}
.card{background:var(--card);border:1px solid var(--bd);border-radius:10px;padding:16px;margin-bottom:14px}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(210px,1fr));gap:12px}
.cake .img{font-size:3rem;text-align:center;background:var(--bg);border-radius:8px;padding:8px;height:90px;display:flex;align-
    items:center;justify-content:center}
.cake .img img{max-height:74px;border-radius:6px}
button.b{background:var(--ac);color:var(--acx);border:0;border-radius:6px;padding:8px 14px;cursor:pointer;font-size:.9rem}
button.a{background:transparent;color:var(--tx);border:1px solid var(--bd)}
button.s{padding:3px 9px;font-size:.8rem}
input,select,textarea{padding:8px;border:1px solid var(--bd);border-radius:6px;background:var(--bg);color:var(--tx);font-
    size:.9rem;max-width:100%}
label{font-size:.85rem;color:var(--mut);display:flex;flex-direction:column;gap:3px}
.row{display:flex;gap:10px;flex-wrap:wrap;align-items:flex-end;margin-bottom:10px}
.tw{overflow-x:auto}
table{border-collapse:collapse;width:100%;font-size:.88rem}
th,td{text-align:left;padding:7px 9px;border-bottom:1px solid var(--bd);white-space:nowrap}
th{color:var(--mut)}
h2{margin:0 0 10px;font-size:1.05rem}h3{margin:6px 0;font-size:1rem}
.bad{color:var(--bad);font-weight:700}.ok{color:var(--ok);font-weight:700}.mut{color:var(--mut);font-size:.85rem}
.price{color:var(--ac);font-weight:700;font-size:1.1rem}
.login{max-width:360px;margin:0 auto}.login input,.login select{width:100%;margin-bottom:8px}
</style>
</head>
<body>
<header><h1>■ Sweet Slice - Online Cake Shop</h1><nav id="nav"></nav></header>
<main id="app"></main>
<script>
const $=id=>document.getElementById(id);
const esc=s=>String(s).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));
const today=()=>new Date().toISOString().slice(0,10);
const add=(d,n)=>{const x=new Date(d);x.setDate(x.getDate()+n);return x.toISOString().slice(0,10)};
const msg=t=>alert(t);
const SIZES={'0.5 kg':0.6,'1 kg':1,'2 kg':1.9};
const STAT=['Placed','Confirmed','Baking','Out for Delivery','Delivered'];
let nid=100;
const D={
 admin:{u:'admin',p:'admin123'},
 users:[{u:'guest',p:'guest123',name:'Anita Rao',email:'anita@mail.com',phone:'9876543210',cart:[],addr:[{id:1,name:'Anita
    Rao',line:'12 Lake View Road',city:'Chennai',pin:'600001',phone:'9876543210'}]}],
 cats:['Birthday','Wedding','Anniversary'],
 cakes:[
  {id:1,name:'Chocolate Truffle',cat:'Birthday',sub:'Chocolate',price:650,desc:'Rich dark chocolate sponge with truffle
    ganache.',img:'■',data:''},
  {id:2,name:'Vanilla Dream',cat:'Birthday',sub:'Vanilla',price:500,desc:'Soft vanilla sponge with whipped
    cream.',img:'■',data:''},
  {id:3,name:'Red Velvet Heart',cat:'Anniversary',sub:'Red Velvet',price:900,desc:'Heart-shaped red velvet with cream cheese
    frosting.',img:'❤■',data:''},
  {id:4,name:'Fruit Delight',cat:'Anniversary',sub:'Fruit',price:750,desc:'Fresh seasonal fruits on a light
    sponge.',img:'■',data:''},
  {id:5,name:'Royal Tier Cake',cat:'Wedding',sub:'Vanilla',price:2500,desc:'Three-tier fondant cake with sugar
    flowers.',img:'■',data:''},
  {id:6,name:'Black Forest',cat:'Birthday',sub:'Chocolate',price:600,desc:'Classic cherry and chocolate
    shavings.',img:'■',data:''},
  {id:7,name:'Butterscotch Crunch',cat:'Birthday',sub:'Butterscotch',price:550,desc:'Butterscotch cream with crunchy
    praline.',img:'■',data:''},
  {id:8,name:'Rose Pistachio',cat:'Wedding',sub:'Fruit',price:2200,desc:'Rose-flavoured sponge with pistachio
    cream.',img:'■',data:''}],
 orders:[]
};
let role=null,me=null,view='login',q={},editC=null,editA=null,pend='';

const go=v=>{view=v;render()};
function render(){
 let n=[];
 if(role=='user')n=[['shop','Cakes'],['cart','Cart ('+me.cart.length+')'],['addr','Shipping Address'],['my','My Orders']];
 if(role=='admin')n=[['prod','Manage Products'],['ord','Manage Orders']];
 n.push(me?['out','Logout ('+esc(me.u)+')']:['login','Login / Register']);
 $('nav').innerHTML=n.map(x=>`<button class="${x[0]==view?'on':''}"
    onclick="${x[0]=='out'?'logout()':`go('${x[0]}')`}">${x[1]}</button>`).join('');
 $('app').innerHTML=({login:loginV,shop:shopV,cart:cartV,addr:addrV,my:myV,prod:prodV,ord:ordV})[view]();
```

```
 if(view=='shop')showList();
}
function logout(){role=me=null;go('login')}
const pic=c=>c.data?`<img src="${c.data}" alt="">`:c.img;
const cake=id=>D.cakes.find(c=>c.id==id);

function loginV(){
 return `<div class="login"><div class="card"><h2>Login</h2><select id="lr"><option value="user">Customer</option><option
    value="admin">Admin</option></select>
 <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password"><button class="b"
    onclick="login()">Login</button>
 <p class="mut">Demo: guest / guest123 | admin / admin123</p></div>
 <div class="card"><h2>Register</h2><input id="rn" placeholder="Full name"><input id="re" placeholder="Email"><input id="rh"
    placeholder="Phone (10 digits)">
 <input id="ru" placeholder="Username"><input id="rp" type="password" placeholder="Password (min 6)"><button class="b"
    onclick="reg()">Register</button></div></div>`;
}
function login(){
 const r=$('lr').value,u=$('lu').value.trim(),p=$('lp').value;
 const f=r=='admin'?(D.admin.u==u&&D.admin.p==p?D.admin:null):D.users.find(x=>x.u==u&&x.p==p);
 if(!f)return msg('Invalid username or password');
 role=r;me=f;go(r=='admin'?'prod':'shop');
}
function reg(){
 const o={name:$('rn').value.trim(),email:$('re').value.trim(),phone:$('rh').value.trim(),u:$('ru').value.trim(),p:$('rp').value
    ,cart:[],addr:[]};
 if(!o.name||!o.u)return msg('Please fill all fields');
 if(!/^\S+@\S+\.\S+$/.test(o.email))return msg('Enter a valid email');
 if(!/^\d{10}$/.test(o.phone))return msg('Phone must be 10 digits');
 if(o.p.length<6)return msg('Password must have at least 6 characters');
 if(D.users.some(x=>x.u==o.u))return msg('Username already taken');
 D.users.push(o);msg('Registered successfully. Please login.');
}

/* ---- customer: search, filter, list, details, cart ---- */
function shopV(){
 const subs=[...new Set(D.cakes.map(c=>c.sub))];
 return `<div class="card"><h2>Search and Filter</h2><div class="row"><label>Search<input id="fq" oninput="showList()"
    placeholder="Cake name"></label>
 <label>Category<select id="fc" onchange="showList()"><option
    value="">All</option>${D.cats.map(c=>`<option>${esc(c)}</option>`).join('')}</select></label>
 <label>Sub category (flavour)<select id="fs" onchange="showList()"><option
    value="">All</option>${subs.map(c=>`<option>${esc(c)}</option>`).join('')}</select></label>
 <label>Min price<input id="fa" type="number" min="0" oninput="showList()" style="width:90px"></label><label>Max price<input
    id="fb" type="number" min="0" oninput="showList()" style="width:90px"></label></div></div><div id="list"></div>`;
}
function showList(){
 const t=$('fq').value.toLowerCase(),c=$('fc').value,s=$('fs').value,a=+$('fa').value||0,b=+$('fb').value||1e9;
 const l=D.cakes.filter(x=>x.name.toLowerCase().includes(t)&&(!c||x.cat==c)&&(!s||x.sub==s)&&x.price>=a&&x.price<=b);
 $('list').innerHTML=`<div class="grid">`+l.map(x=>`<div class="card cake"><div
    class="img">${pic(x)}</div><h3>${esc(x.name)}</h3>
 <div class="mut">${esc(x.cat)} | ${esc(x.sub)}</div><div class="price">Rs. ${x.price} <span class="mut">/ 1 kg</span></div><p
    class="mut">${esc(x.desc)}</p>
 <label>Size<select id="sz${x.id}">${Object.keys(SIZES).map(k=>`<option ${k=='1
    kg'?'selected':''}>${k}</option>`).join('')}</select></label>
 <label>Message on cake<input id="ms${x.id}" maxlength="30" placeholder="e.g. Happy Birthday Ravi"></label><br>
 <button class="b s" onclick="addCart(${x.id})">Add to Cart</button></div>`).join('')+'</div>'+(l.length?'':'<div class="card
    mut">No cakes match your filters.</div>');
}
function addCart(id){
 const c=cake(id),sz=$('sz'+id).value;
 me.cart.push({id,name:c.name,size:sz,msg:$('ms'+id).value.trim(),price:Math.round(c.price*SIZES[sz])});
 msg(c.name+' ('+sz+') added to cart');render();
}
function cartV(){
 const t=me.cart.reduce((s,i)=>s+i.price,0);
 if(!me.cart.length)return '<div class="card">Your cart is empty.</div>';
 return `<div class="card tw"><h2>My
    Cart</h2><table><tr><th>Cake</th><th>Size</th><th>Message</th><th>Price</th><th></th></tr>`+
 me.cart.map((i,k)=>`<tr><td>${esc(i.name)}</td><td>${i.size}</td><td>${esc(i.msg)||'-'}</td><td>Rs. ${i.price}</td><td><button
    class="b a s" onclick="me.cart.splice(${k},1);render()">Remove</button></td></tr>`).join('')+
 `</table><p><b>Total: Rs. ${t}</b></p></div>
 <div class="card"><h2>Place Order</h2><div class="row"><label>Shipping address<select id="oa">${me.addr.map(a=>`<option
    value="${a.id}">${esc(a.line)}, ${esc(a.city)} - ${a.pin}</option>`).join('')}</select></label>
 <label>Delivery date<input type="date" id="od" min="${add(today(),1)}" value="${add(today(),2)}"></label>
 <label>Payment<select id="op"><option>Cash on
    Delivery</option><option>UPI</option><option>Card</option></select></label><button class="b" onclick="place()">Place
    Order</button></div>
 ${me.addr.length?'':'<p class="bad">Please add a shipping address first.</p>'}</div>`;
}
function place(){
 const a=me.addr.find(x=>x.id==$('oa').value);
 if(!a)return msg('Add a shipping address first (Shipping Address menu)');
 if($('od').value<=today())return msg('Delivery date must be after today');
 const o={id:++nid,u:me.u,items:me.cart.slice(),total:me.cart.reduce((s,i)=>s+i.price,0),addr:`${a.name}, ${a.line}, ${a.city} -
     ${a.pin}, Ph ${a.phone}`,pay:$('op').value,dd:$('od').value,st:'Placed',loc:'Shop - Order received',date:today()};
 D.orders.push(o);me.cart=[];msg('Order #'+o.id+' placed! Confirmation email sent to '+me.email+'.');go('my');
}
function addrV(){
 const e=me.addr.find(a=>a.id==editA)||{name:me.name,line:'',city:'',pin:'',phone:me.phone};
 return `<div class="card"><h2>${editA?'Update':'Add'} Shipping Address</h2><div class="row"><label>Name<input id="an"
    value="${esc(e.name)}"></label><label>Address<input id="al" value="${esc(e.line)}"></label>
 <label>City<input id="ac" value="${esc(e.city)}"></label><label>PIN<input id="ap" value="${e.pin}"
```

```
    style="width:90px"></label><label>Phone<input id="ah" value="${e.phone}"></label>
 <button class="b" onclick="saveAddr()">Save</button>${editA?'<button class="b a"
    onclick="editA=null;render()">Cancel</button>':''}</div></div>
 <div class="card tw"><h2>My
    Addresses</h2><table><tr><th>Name</th><th>Address</th><th>City</th><th>PIN</th><th>Phone</th><th></th></tr>`+
 (me.addr.map(a=>`<tr><td>${esc(a.name)}</td><td>${esc(a.line)}</td><td>${esc(a.city)}</td><td>${a.pin}</td><td>${a.phone}</td><
    td><button class="b a s" onclick="editA=${a.id};render()">Edit</button> <button class="b a s"
    onclick="me.addr=me.addr.filter(x=>x.id!=${a.id});render()">Delete</button></td></tr>`).join('')||'<tr><td colspan="6"
    class="mut">No address saved.</td></tr>')+'</table></div>';
}
function saveAddr(){
 const o={name:$('an').value.trim(),line:$('al').value.trim(),city:$('ac').value.trim(),pin:$('ap').value.trim(),phone:$('ah').v
    alue.trim()};
 if(!o.name||!o.line||!o.city)return msg('Fill name, address and city');
 if(!/^\d{6}$/.test(o.pin))return msg('PIN must be 6 digits');
 if(!/^\d{10}$/.test(o.phone))return msg('Phone must be 10 digits');
 if(editA)Object.assign(me.addr.find(a=>a.id==editA),o);else me.addr.push({id:++nid,...o});
 editA=null;render();
}
function myV(){
 return `<div class="card tw"><h2>My Orders and Shipping
    Status</h2><table><tr><th>Order</th><th>Items</th><th>Total</th><th>Delivery date</th><th>Status</th><th>Shipping
    location</th></tr>`+
 (D.orders.filter(o=>o.u==me.u).map(o=>`<tr><td>#${o.id}</td><td>${o.items.map(i=>esc(i.name)+' ('+i.size+')').join(',
    ')}</td><td>Rs. ${o.total}</td><td>${o.dd}</td>
 <td class="${o.st=='Delivered'?'ok':''}">${o.st}</td><td>${esc(o.loc)}</td></tr>`).join('')||'<tr><td colspan="6"
    class="mut">No orders yet.</td></tr>')+'</table></div>';
}

/* ---- admin ---- */
function prodV(){
 const e=cake(editC)||{name:'',cat:D.cats[0],sub:'',price:'',desc:'',img:'■'};
 return `<div class="card"><h2>${editC?'Update':'Post'} Cake Details</h2><div class="row"><label>Product name<input id="pn"
    value="${esc(e.name)}"></label>
 <label>Category<select id="pc">${D.cats.map(c=>`<option
    ${c==e.cat?'selected':''}>${esc(c)}</option>`).join('')}</select></label>
 <label>Sub category (flavour)<input id="ps" value="${esc(e.sub)}"></label><label>Price / kg (Rs.)<input id="pp" type="number"
    min="1" value="${e.price}" style="width:100px"></label>
 <label>Description<input id="pd" value="${esc(e.desc)}"></label><label>Emoji icon<input id="pe" value="${e.img}"
    style="width:70px"></label>
 <label>Upload image<input type="file" id="pf" accept="image/*" onchange="upl(this)"></label>
 <button class="b" onclick="saveCake()">Save</button>${editC?'<button class="b a"
    onclick="editC=null;render()">Cancel</button>':''}</div></div>
 <div class="card tw"><h2>Products</h2><table><tr><th>Image</th><th>Name</th><th>Category</th><th>Flavour</th><th>Price /
    kg</th><th></th></tr>`+
 D.cakes.map(c=>`<tr><td style="font-size:1.5rem">${c.data?`<img src="${c.data}" width="34"
    alt="">`:c.img}</td><td>${esc(c.name)}</td><td>${esc(c.cat)}</td><td>${esc(c.sub)}</td><td>Rs. ${c.price}</td>
 <td><button class="b a s" onclick="editC=${c.id};render()">Edit</button> <button class="b a s"
    onclick="D.cakes=D.cakes.filter(x=>x.id!=${c.id});render()">Delete</button></td></tr>`).join('')+'</table></div>';
}
function upl(f){const r=new FileReader();r.onload=()=>pend=r.result;if(f.files[0])r.readAsDataURL(f.files[0])}
function saveCake(){
 const o={name:$('pn').value.trim(),cat:$('pc').value,sub:$('ps').value.trim(),price:+$('pp').value,desc:$('pd').value.trim(),im
    g:$('pe').value||'■'};
 if(!o.name||!o.sub||o.price<1)return msg('Enter name, flavour and price');
 if(editC){const c=cake(editC);Object.assign(c,o);if(pend)c.data=pend}else D.cakes.push({id:++nid,...o,data:pend});
 pend='';editC=null;render();
}
function ordV(){
 return `<div class="card tw"><h2>Manage
    Orders</h2><table><tr><th>Order</th><th>Customer</th><th>Items</th><th>Total</th><th>Deliver
    to</th><th>Status</th><th>Shipping location</th><th></th></tr>`+
 (D.orders.map(o=>`<tr><td>#${o.id}</td><td>${esc(o.u)}</td><td>${o.items.map(i=>esc(i.name)+' ('+i.size+')'+(i.msg?'
    "'+esc(i.msg)+'"':'')).join(', ')}</td><td>Rs. ${o.total}</td><td>${esc(o.addr)}<br><span class="mut">${o.dd},
    ${o.pay}</span></td>
 <td><select id="st${o.id}">${STAT.map(s=>`<option ${s==o.st?'selected':''}>${s}</option>`).join('')}</select></td><td><input
    id="lc${o.id}" value="${esc(o.loc)}"></td>
 <td><button class="b s" onclick="updO(${o.id})">Update</button></td></tr>`).join('')||'<tr><td colspan="8" class="mut">No
    orders yet.</td></tr>')+'</table></div>';
}
function updO(id){const o=D.orders.find(x=>x.id==id);o.st=$('st'+id).value;o.loc=$('lc'+id).value;msg('Order #'+id+'
    updated');render()}
render();
</script>
</body>
</html>
```

Output

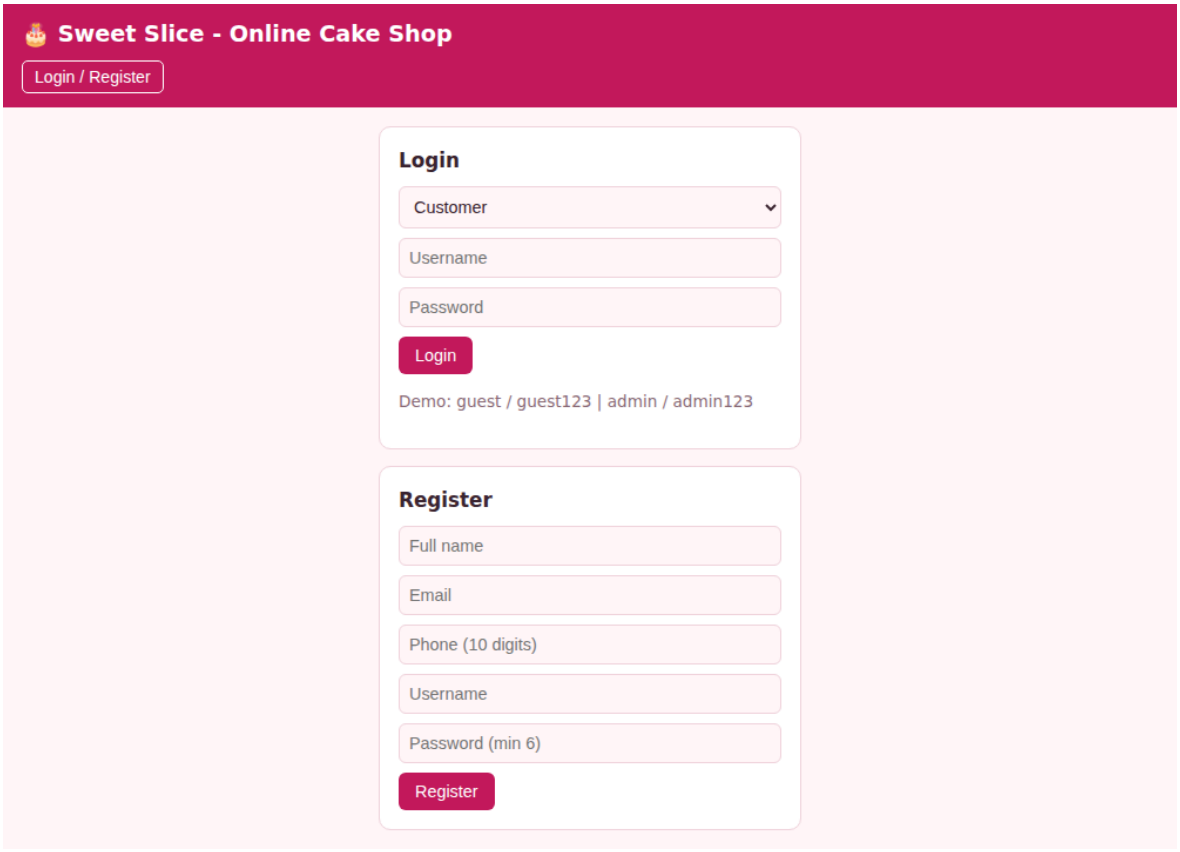


Fig 1: Login and registration page

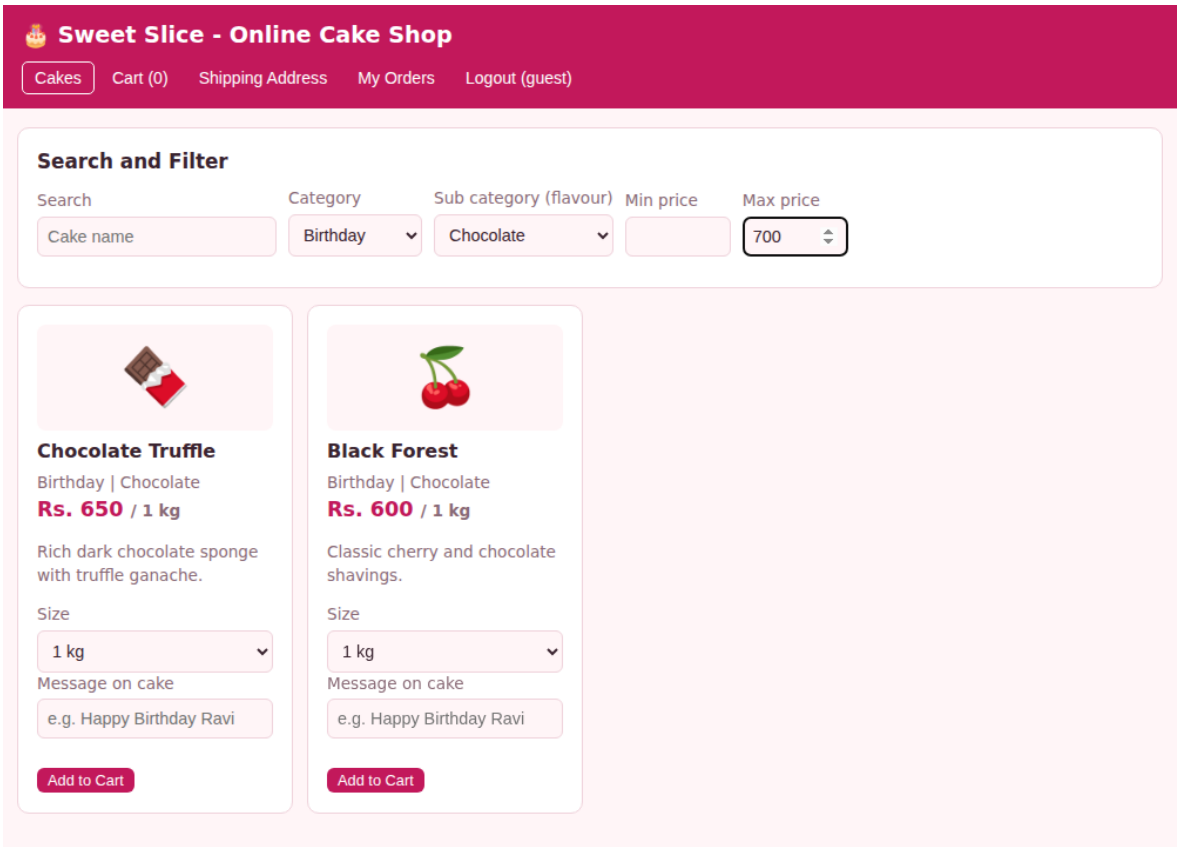


Fig 2: Customer - Search and filter (Birthday, Chocolate, up to Rs. 700)

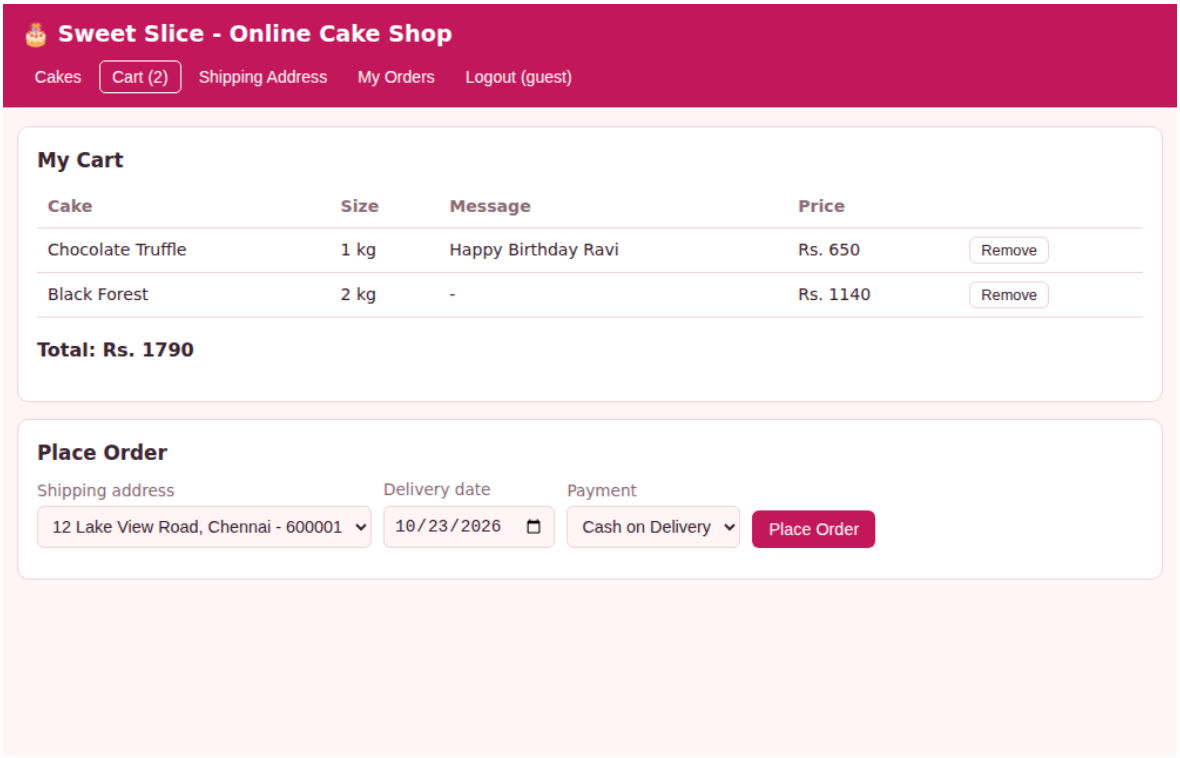


Fig 3: Customer - Cart with sizes and messages

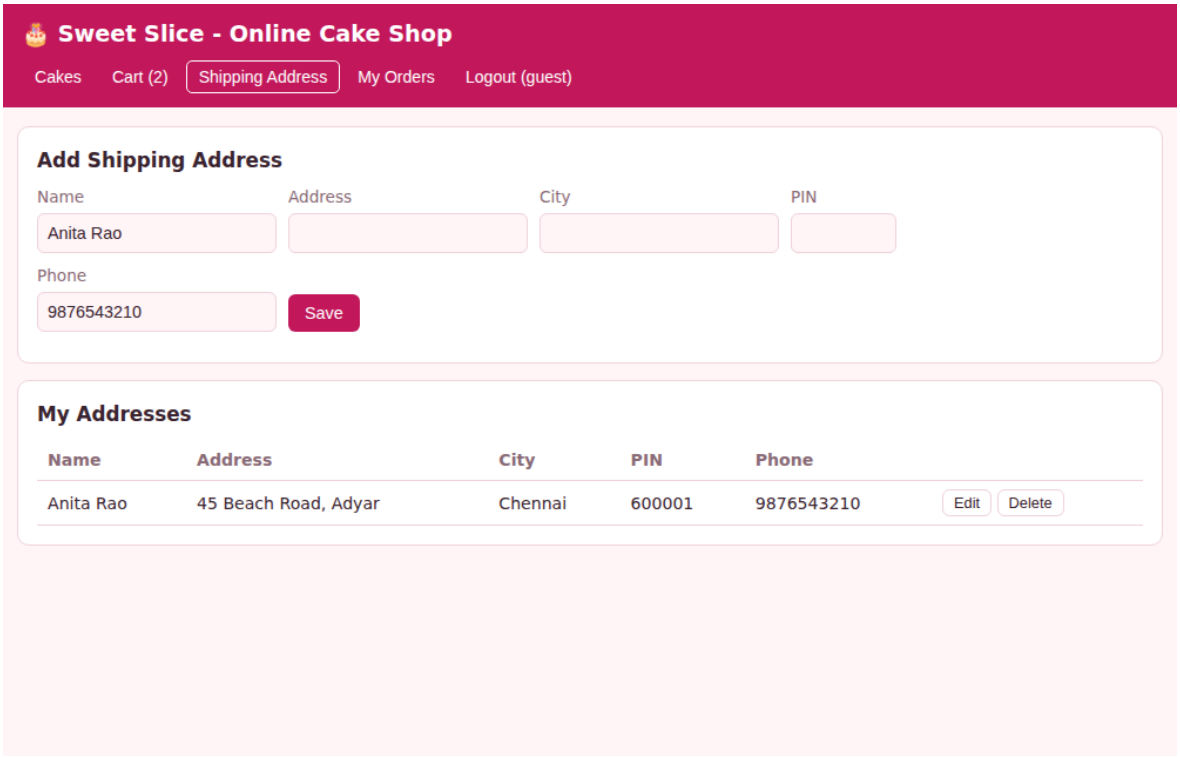


Fig 4: Customer - Update shipping address

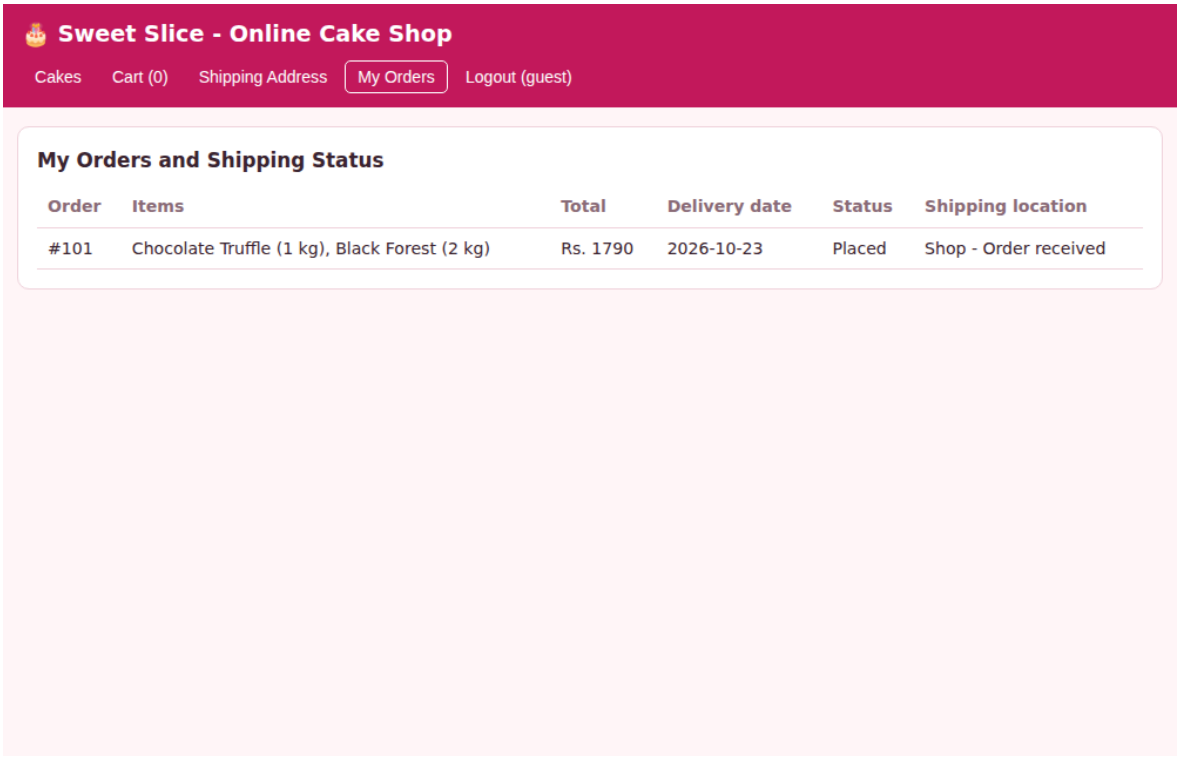


Fig 5: Customer - Order placed (order #101)

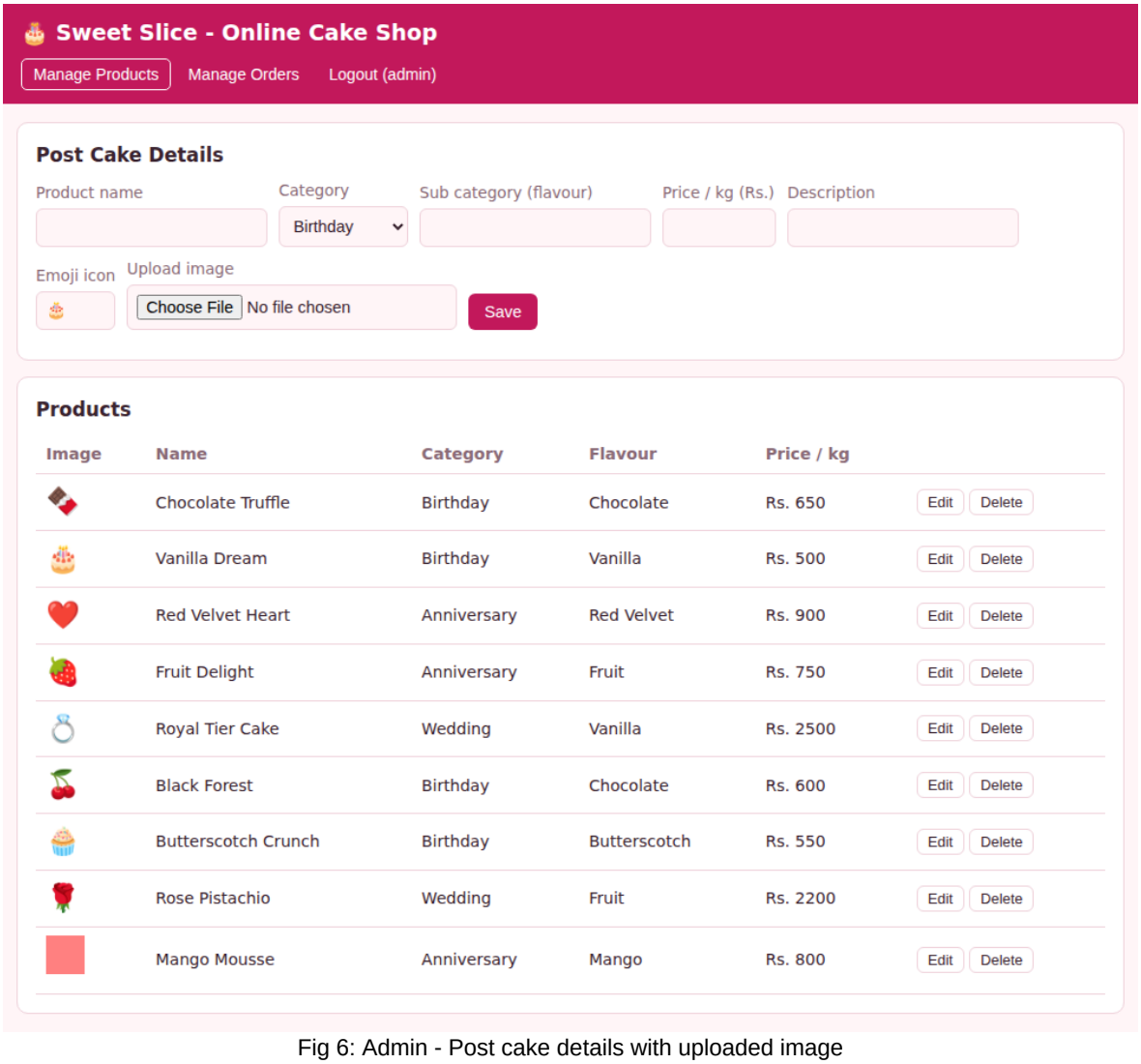


Fig 6: Admin - Post cake details with uploaded image

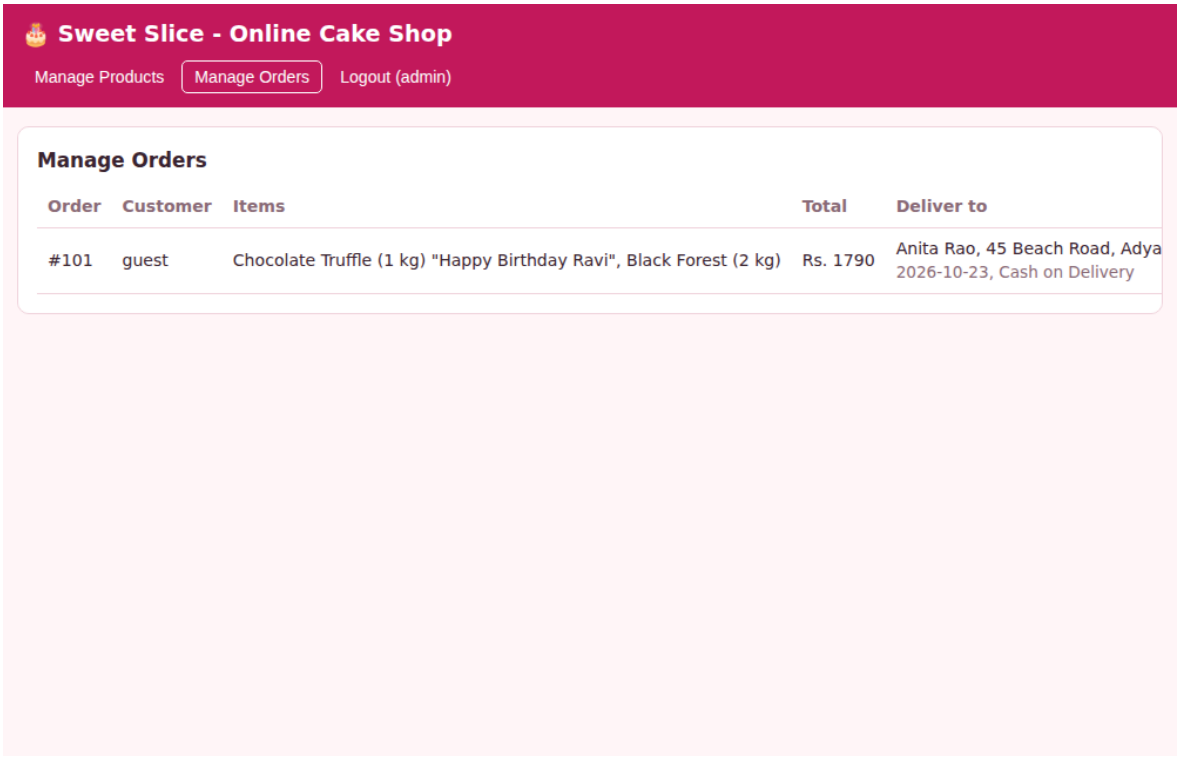


Fig 7: Admin - Manage orders, status and shipping location

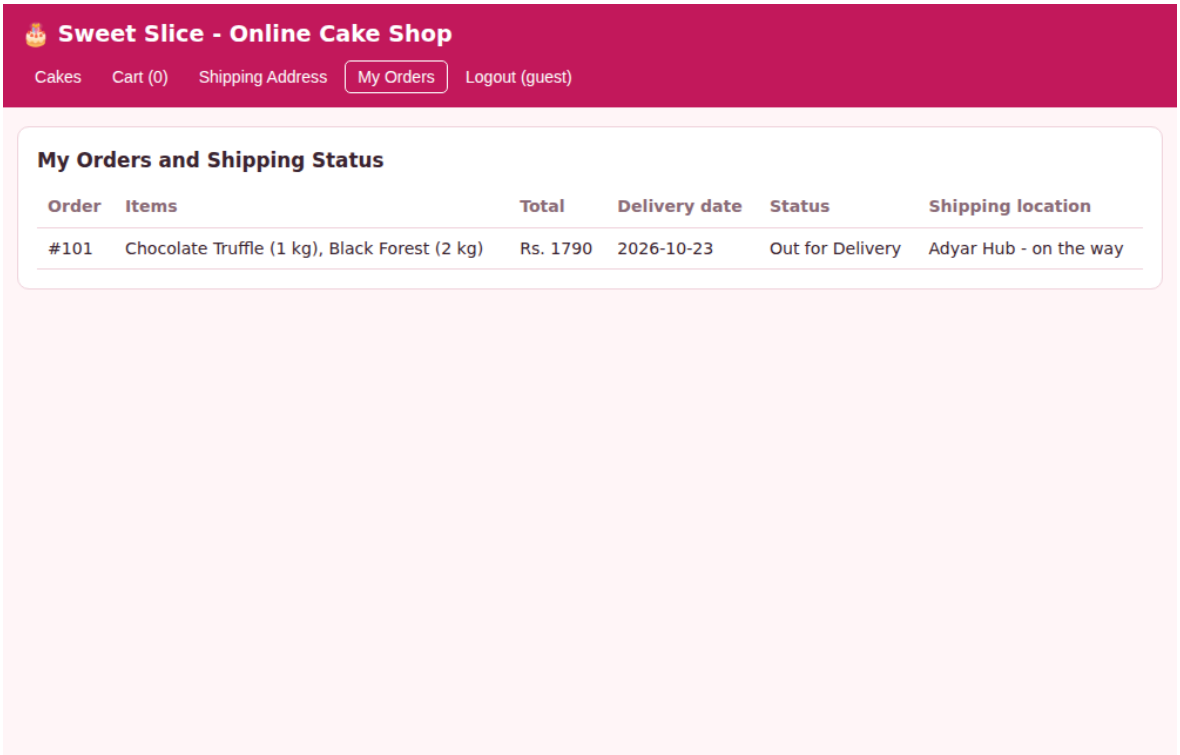


Fig 8: Customer - Shipping status of the order

Result

The online cake ordering website was developed using HTML, CSS and JavaScript. Customer operations (registration, login, search and filter, cart, address, order, shipping status) and admin operations (product posting with images, order management, status and shipping location update) were executed successfully.